Personalizing POI Descriptions with LLMs: A Study of User Perceptions and Informational Value

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Abstract

Personalization is essential for enhancing user engagement with travel-related content. This study investigates the effectiveness of AI-generated descriptions for Points of Interest (POIs), including hotels, restaurants, museums, and gardens, by adapting content to better align with individual user preferences. A prototype system was developed, allowing users to input their preferences and interests, after which they were presented with both original and LLM-adapted POI descriptions. Participants were then asked to rate the descriptions based on their alignment with personal interests and familiarity with the locations. The study aims to evaluate the impact of personalized descriptions on user acceptance and perceived relevance, offering insights into the potential of large language models for enhancing tourism and travel experiences. The findings reveal key factors influencing user preferences and provide recommendations for improving AI-driven content personalization strategies.

CCS Concepts

• **Human-centered computing** → **User centered design**; *User studies.*

Keywords

poi adaptation, LLM-generated content, ai-driven descriptions, content personalization

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1 Introduction

Personalization has become a key factor in improving user engagement in various digital experiences, including travel and tourism. As travellers seek more relevant and tailored recommendations, AI advancements, particularly large language models (LLM), presents new opportunities to adapt content to align with individual preferences dynamically. One such application is personalized descriptions of Points of Interest (POIs), such as hotels, restaurants, museums, and gardens. Although traditional POI descriptions are often static and generalized, AI-driven personalization can refine these descriptions to match user-specific interests, past experiences, and contextual relevance, enhancing both engagement and comprehension.

Much prior research has focused on POI recommendation and ranking systems, where AI models suggest locations based on user preferences, behaviors, and contextual factors. However, less attention has been given to how the content of each POI itself can be adapted to better engage users. When individuals access POI descriptions, the same information is typically presented to all users, despite variations in their backgrounds, interests, and motivations to visit a place. For example, a museum description could emphasize historical significance for a history enthusiast while focusing on interactive exhibits for a family traveler. Generative AI and LLMs provide a unique opportunity to dynamically tailor content without altering its factual meaning, ensuring that the description remains relevant and appealing to different audiences.

To explore this potential, we developed a prototype system that allows users to enter their preferences and interests and then dynamically presents them with two versions of each POI description: the original and an LLM-adapted version. Participants were asked to evaluate the descriptions based on alignment with their preferences, perceived appeal, and familiarity with the location. The key objective is to assess whether AI-generated adaptations are acceptable and meaningful, effectively convey the original value of POI, and improve user engagement.

This study contributes to the growing field of AI-powered content personalization in tourism by investigating the role of LLMs in dynamic adaptation of textual content. By analyzing participant feedback, we aim to provide insight into how AI-generated descriptions can improve engagement while preserving accuracy and authenticity.

The remainder of this paper is structured as follows. Section 2 details our methodology, including the experimental setup, data collection process, and evaluation metrics. Section 3 presents the results and analysis of user perceptions across significance, trustworthiness, and clarity dimensions. Section 4 discusses the implications of our findings and limitations of the study. Section 5 outlines directions for future research, and Section 6 concludes with key takeaways and practical implications for tourism platforms.

2 Methodology

To assess the effectiveness of LLM-generated adaptations of POI descriptions, we conducted a user study where participants evaluated both original and AI-adapted descriptions based on their alignment with personal preferences, perceived appeal, and ability to convey the original value of the POI. This section details the study design, participant recruitment, evaluation criteria.

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2.1 Study Design

The study was structured as a within-subjects experiment, where each participant evaluated multiple POIs and compared both original and AI-adapted descriptions. This approach allowed for direct comparisons between static and personalized content while minimizing individual biases. The study included ten POIs, categorized into five different types—hotels, restaurants, museums, gardens, and landmarks—ensuring a diverse representation of travel-related locations. Each category contained two POIs, carefully selected to provide a balanced mix of well-known and lesser-known places.

For each POI, participants were shown two descriptions side by side, but they were not informed which one was AI-adapted and which one was the original. One description was the original, unmodified version, while the other was dynamically generated using an LLM, tailored to the participant's previously provided interests and preferences. To reduce order effects, the position of the AI-adapted and original descriptions was randomized for each POI. The study was designed to assess whether AI-generated adaptations were both meaningful and engaging while preserving the original value of the POI. It also examined whether personalized descriptions increased user interest in visiting the locations.

2.2 Procedure

The study began with a consent form, where participants were informed about the purpose of the study, the voluntary nature of participation, and data confidentiality. After providing consent, participants proceeded to a background and preference survey. This survey collected demographic information such as age, gender, nationality, profession, and current city. Additionally, participants specified their travel-related preferences, selecting from a list of hobbies, preferred travel styles, and key tourism interests. This input was later used to personalize POI descriptions for the evaluation phase.

Once the survey was completed, participants moved on to the main evaluation task, where they were presented with ten POIs, each accompanied by an image. For each POI, they reviewed two descriptions (one original, one AI-adapted) without knowing which was which. Participants were then asked to evaluate each description individually based on clarity, trustworthiness, and effectiveness in conveying the significance of the POI.

Following the individual assessment, participants completed a comparative evaluation, where they indicated which description was more engaging, informative, appealing, and likely to encourage them to visit the location. To further understand how prior knowledge influenced perception, participants were also asked to report their familiarity with each POI, specifying whether they had previously visited it, heard about it, or were entirely unfamiliar with it.

Upon completing evaluations for all ten POIs, participants submitted their responses, concluding the study. The entire process took approximately 20 to 30 minutes, with minor variations depending on reading speed and response time. All data was collected anonymously, ensuring participant privacy and compliance with ethical research standards.

2.3 Participants

The study recruited 51 participants through email and direct messages to friends, family, and colleagues. The participant pool was designed to include a diverse range of individuals with varying backgrounds and travel preferences to ensure the findings would be broadly applicable. The age of participants ranged from 22 to 49 years (M = 30.94, SD = 6.74). The sample comprised 34 males and 17 females, with diverse nationalities including 28 Indian and 21 German participants, along with Chinese and Iranian representation. Regarding marital status, 25 were married, 22 single, 3 in a relationship, and 1 divorced. The professional background was predominantly students (22) and engineers (16), with a smaller number of academics, healthcare professionals, and public service workers. Travel experience varied across the sample, with most participants describing themselves as intermediate (21) or experienced (19) travelers, while 9 identified as beginners and 2 as experts. Participants were required to be at least 18 years old and have a basic level of English proficiency to ensure they could effectively engage with the POI descriptions. No specific travel experience was required, as the study aimed to assess how AI-generated descriptions appeal to both frequent and occasional travelers.

3 Results

This section presents the findings from our user study evaluating LLM-adapted POI descriptions against original manual descriptions. We analyze participant responses across three core metrics: significance, trustworthiness, and clarity, followed by comparative preference evaluations and overall feedback on the adaptive personalization concept. The study collected 510 paired observations (51 participants × 10 POIs) for each metric, enabling robust statistical analysis of user perceptions. Table 1 provides an overview of the rating distributions across all three evaluation dimensions.

Table 1: Summary of rating distributions for manual and AI-generated descriptions across three evaluation metrics (n=510 per condition)

Metric	Rating	Manual	AI
Significance	Strongly Disagree	7	8
	Disagree	21	24
	Neutral	138	118
	Agree	281	276
	Strongly Agree	63	84
Trustworthiness	Not at all	8	4
	Slightly	43	33
	Moderately	170	140
	Very	235	261
	Extremely	54	72
Clarity	Very Unclear	3	5
	Unclear	33	24
	Neutral	147	109
	Clear	260	284
	Very Clear	67	88

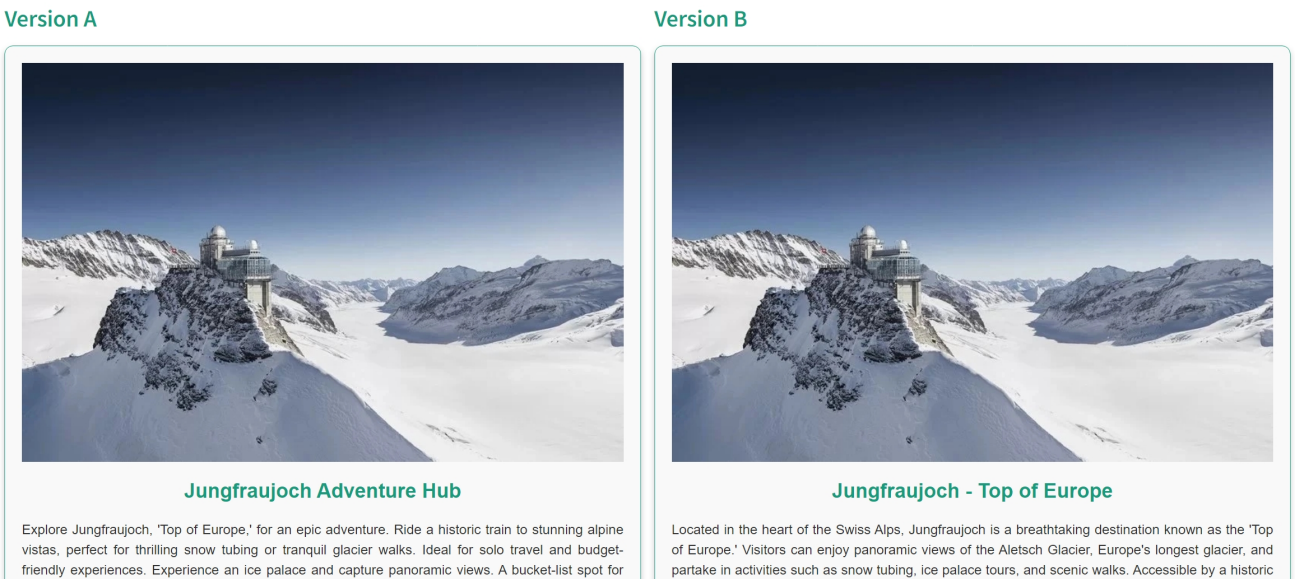

Figure 1: Two versions of POIs as shown to users

3.1 Perceived Significance

Participants rated both description versions on how effectively they communicated the significance and offerings of each POI using a 5-point Likert scale ranging from Strongly Disagree to Strongly Agree. Figure 2 illustrates the distribution of responses for manual and AI-generated descriptions.

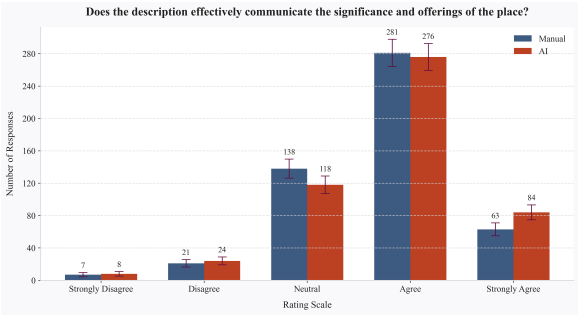

Figure 2: Comparison of perceived significance between manual and AI-generated descriptions. Error bars represent Poisson standard error estimates ($\sqrt{n}$).

The results show that both manual and AI descriptions received predominantly positive ratings, with the majority of responses falling in the Agree category (Manual: 281, AI: 276). However, AI-generated descriptions demonstrated a notable shift toward stronger agreement, with 84 Strongly Agree responses compared to 63 for manual descriptions. This increase was primarily driven by a reduction in Neutral responses (Manual: 138, AI: 118), suggesting that personalized descriptions moved participants from ambivalence toward stronger endorsement. A chi-square test revealed no statistically significant difference between the distributions ($\chi^2(4) = 4.87$, $p = 0.3005$), indicating that AI adaptations maintained equivalence with manual descriptions while showing a trend toward higher top-box scores.

3.2 Trustworthiness

Trustworthiness was assessed on a 5-point scale ranging from Not at all to Extremely. Figure 3 presents the comparative distribution of trust ratings.

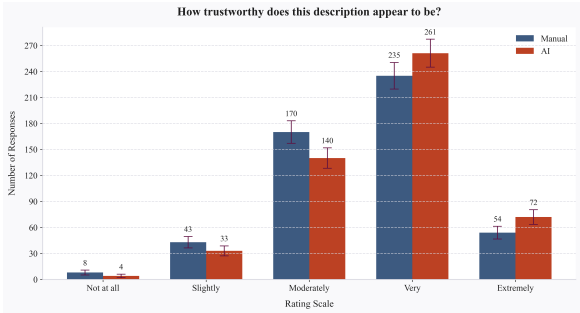

Figure 3: Comparison of perceived trustworthiness between manual and AI-generated descriptions. Error bars represent Poisson standard error estimates ($\sqrt{n}$).

AI-generated descriptions received notably higher trust ratings than manual descriptions. The most striking difference appeared at the highest end of the scale, with AI descriptions receiving 72 Extremely trustworthy ratings compared to 54 for manual descriptions. Additionally, AI descriptions garnered 261 Very trustworthy

ratings versus 235 for manual versions. At the lower end of the scale, AI descriptions received only 4 Not at all ratings compared to 8 for manual descriptions, representing a 50% reduction in the most skeptical responses. The shift from Moderate trust toward Very and Extremely trust was particularly pronounced. A chi-square test indicated marginal statistical significance ($\chi^2(4) = 9.49$, $p = 0.050$), suggesting that AI personalization had a meaningful positive impact on perceived trustworthiness.

3.3 Clarity and Completeness

Participants evaluated how clear and complete the information was using a 5-point scale from Very Unclear to Very Clear. Figure 4 shows the distribution of clarity ratings.

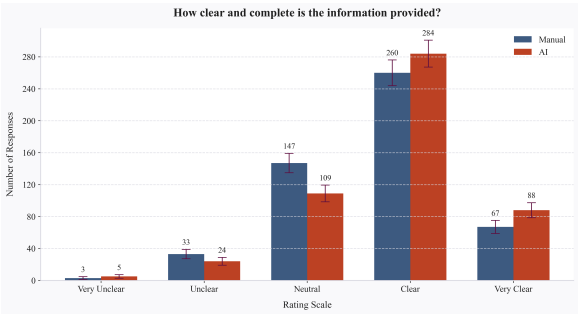

Figure 4: Comparison of perceived clarity and completeness between manual and AI-generated descriptions. Error bars represent Poisson standard error estimates ($\sqrt{n}$).

AI-generated descriptions demonstrated superior clarity ratings compared to manual versions. Clear ratings increased from 260 for manual descriptions to 284 for AI descriptions, while Very Clear ratings rose from 67 to 88. This improvement was accompanied by a substantial reduction in Neutral responses (from 147 to 109), indicating that AI adaptations helped participants form more definitive judgments about information quality. The Unclear category also decreased from 33 to 24 responses. A chi-square test confirmed statistical significance ($\chi^2(4) = 11.47$, $p = 0.0218$), providing evidence that personalized AI descriptions significantly enhanced perceived clarity and completeness of information.

3.4 Comparative Preferences

Beyond individual ratings, participants indicated which description version they found more engaging, relevant, and appealing. For the engaging preference, responses were nearly evenly split between AI-generated (221 preferences) and manual (220 preferences), with 69 participants indicating both versions were equally engaging. This near-parity suggests that AI personalization successfully matched the engagement level of carefully crafted manual descriptions while adding the benefit of customization.

3.5 Overall Study Feedback

Following the POI evaluations, a subset of 17 participants provided overall ratings on three dimensions using a 5-point scale (1 = Poor, 5 = Excellent). The overall experience with POI descriptions received a mean rating of 3.88 (SD = 0.70, Median = 4.0), indicating generally positive reception. Participants expressed moderate to strong support for the concept of automatically adapting POI descriptions based on user interests, with a mean rating of 3.75 (SD = 0.93, Median = 4.0). Comfort with reading AI-generated descriptions when planning visits to new places received the highest mean rating of 3.94 (SD = 0.83, Median = 4.0), suggesting participants were receptive to AI-generated travel content when it demonstrated quality and relevance. The relatively low standard deviations (all below 1.0) indicate consensus among participants without extreme polarization of opinions.

4 Discussion

This study investigated whether LLM-generated personalization of POI descriptions could enhance user engagement and perceived information quality while maintaining trustworthiness and accuracy. Our findings demonstrate that AI-adapted descriptions successfully achieved these objectives across multiple evaluation dimensions, with particularly notable improvements in clarity and trustworthiness.

4.1 Effectiveness of AI Personalization

The results reveal that AI-generated adaptations consistently shifted participant responses from neutral positions toward positive endorsements. Across all three core metrics (significance, trustworthiness, and clarity), we observed a compression of the neutral category and expansion of the top-box ratings. This pattern suggests that personalized descriptions help users form more confident and favorable assessments of POI value. The statistical significance achieved for clarity ($p = 0.0218$) and marginal significance for trustworthiness ($p = 0.050$) provide empirical support for the effectiveness of LLM-based personalization in travel content.

The near-parity in engagement preferences (221 AI vs. 220 manual) is particularly encouraging. It indicates that AI personalization does not merely optimize for a single dimension at the expense of others but rather maintains overall engagement quality while adding customization benefits. This balanced performance is critical for practical deployment, as travel platforms must ensure that personalized content remains compelling across diverse user segments.

4.2 Trust as a Critical Factor

The improvement in perceived trustworthiness warrants special attention. Trust is fundamental to POI recommendations, as users must believe the information is accurate and reliable before making travel decisions. The 50% reduction in Not at all trustworthy ratings and the substantial increase in Extremely trustworthy ratings suggest that personalization, when executed thoughtfully, can enhance rather than undermine credibility. This finding challenges potential concerns that AI-generated content might be perceived as less authentic or reliable than human-authored descriptions.

We hypothesize that this trust enhancement stems from the alignment between description content and user interests. When information resonates with a user's stated preferences and travel style, it may be perceived as more relevant and therefore more

credible. Additionally, the LLM's ability to maintain factual consistency while adjusting emphasis and framing likely contributed to maintaining high trust levels.

4.3 Clarity Through Personalization

The statistically significant improvement in clarity represents one of the study's most compelling findings. By adapting content to match user interests and expertise levels, AI-generated descriptions appear to communicate information more effectively. This suggests that generic, one-size-fits-all descriptions may inadvertently obscure key information for certain user segments. Personalization addresses this limitation by surfacing the most relevant details for each individual.

For example, a museum description personalized for a history enthusiast might emphasize historical context and artifacts, while the same POI described for a family traveler might highlight interactive exhibits and educational programs. Both versions convey accurate information but differ in emphasis and framing, resulting in clearer communication of value propositions aligned with user motivations.

4.4 Participant Demographics and Generalizability

Our participant pool consisted primarily of young, educated individuals (mean age 30.94 years) with intermediate to experienced travel backgrounds. The predominance of students and engineers suggests comfort with technology and potentially higher receptiveness to AI-generated content. These demographic characteristics may have contributed to the positive reception of personalized descriptions and high comfort levels with AI-authored travel content (mean = 3.94).

However, this demographic concentration also limits generalizability. Older travelers, those less familiar with AI technologies, or individuals from different cultural backgrounds might exhibit different preferences and trust patterns. The strong representation of Indian and German participants provides some international diversity but does not capture the full spectrum of global travel audiences.

4.5 Implications for Tourism Platforms

These findings have practical implications for travel and tourism platforms seeking to enhance user engagement. The results suggest that LLM-based personalization can be deployed to adapt POI descriptions dynamically without sacrificing trust or perceived information quality. Platforms could collect user preferences during onboarding or infer them from browsing behavior, then use LLMs to generate tailored descriptions that highlight aspects most relevant to each user.

The high comfort level with AI-generated content (mean = 3.94) indicates that transparency about AI involvement may not be a barrier to adoption, particularly among tech-savvy user segments. However, platforms should consider how to communicate the personalization process to maintain trust and set appropriate expectations.

4.6 Limitations

Several limitations should be acknowledged. First, participants evaluated descriptions in a controlled study environment rather than during actual trip planning, which may not fully capture real-world decision-making contexts. Second, the study did not track whether personalized descriptions influenced actual visit intentions or behaviors beyond stated preferences. Third, the within-subjects design, while enabling direct comparisons, may have introduced demand characteristics or heightened attention to differences between versions. Fourth, the study did not examine potential negative effects of over-personalization, such as filter bubbles or reduced serendipitous discovery of POIs outside user-stated interests.

5 Future Work

Building on the promising results of this study, several research directions warrant further investigation to advance our understanding of LLM-based POI personalization and its practical deployment in travel platforms.

Diverse User Populations. Future studies should recruit participants with greater demographic diversity, particularly including older adults, non-technical professionals, and individuals from varied cultural backgrounds. Understanding how different age groups and cultural contexts influence receptiveness to AI-generated travel content would provide crucial insights for global platform deployment. Additionally, examining how travel expertise level moderates preferences for personalized versus generic descriptions could inform adaptive personalization strategies.

Longitudinal Evaluation. This study captured participant perceptions at a single point in time. Longitudinal research tracking how user preferences and trust in AI-generated descriptions evolve with repeated exposure would illuminate whether positive initial impressions persist or fade over time. Such studies could also investigate whether users develop preferences for specific personalization styles or become more discerning about AI-generated content quality with experience.

Behavioral Outcomes. While our study measured perceptions and stated preferences, future research should examine whether personalized descriptions influence actual behavior. Do personalized POI descriptions increase visit likelihood, booking rates, or satisfaction with chosen destinations? Field studies conducted in partnership with travel platforms could track conversion metrics and post-visit satisfaction to assess real-world impact.

Personalization Granularity. This study implemented personalization at a relatively coarse level based on stated interests and travel styles. Future work could explore finer-grained personalization considering factors such as trip purpose (business vs. leisure), travel companions (solo, couple, family), budget constraints, accessibility needs, and contextual factors like time of year or local events. Investigating optimal personalization granularity would help balance customization benefits against complexity and potential privacy concerns.

Transparency and Explainability. Our study did not inform participants which descriptions were AI-generated during the evaluation phase. Future research should examine how transparency about AI involvement affects user perceptions and preferences. Additionally, providing explanations for why specific aspects were

emphasized in personalized descriptions might enhance trust and user understanding of the personalization process.

Comparison with Recommendation Systems. While this study focused on adapting descriptions of given POIs, future work could integrate personalized descriptions with POI recommendation systems. Investigating how personalized descriptions complement personalized recommendations could optimize the entire discovery-to-booking journey.

Mitigation of Filter Bubbles. Future research should examine potential negative effects of over-personalization, such as filter bubbles that limit serendipitous discovery of POIs outside user-stated interests. Developing strategies to balance personalization with diversity and novelty would ensure that adaptive systems enhance rather than constrain travel exploration.

6 Conclusion

This study demonstrates that large language models can effectively personalize POI descriptions to enhance user engagement and perceived information quality. Through a controlled user study with 51 participants evaluating 510 paired comparisons across 10 diverse POIs, we found that AI-generated adaptations significantly improved perceived clarity ($p = 0.0218$) and showed marginal improvements in trustworthiness ($p = 0.050$) compared to original manual descriptions. AI-personalized descriptions shifted user responses from neutral assessments toward stronger positive endorsements across all evaluated dimensions, while maintaining engagement levels comparable to carefully crafted manual content.

The findings reveal three key insights. First, personalization helps users form more confident and favorable assessments of POI value by surfacing information aligned with their interests and travel styles. Second, thoughtfully executed AI personalization can enhance rather than undermine perceived trustworthiness, challenging concerns about credibility of generated content. Third, participants expressed high comfort with AI-generated travel content (mean = 3.94 on a 5-point scale), suggesting receptiveness to this technology in tourism contexts.

These results have practical implications for travel platforms seeking to scale personalized experiences. LLM-based description adaptation offers a path to deliver customized content without the resource constraints of manual authoring for diverse user segments. However, deployment should consider demographic differences in receptiveness to AI content and implement strategies to prevent over-personalization from limiting serendipitous discovery.

This research contributes to the growing understanding of AI-powered personalization in tourism by demonstrating that content adaptation, not just recommendation, plays a valuable role in enhancing user experience. As large language models continue to advance, their application to travel content personalization presents opportunities to make tourism information more accessible, relevant, and engaging for diverse global audiences.

References

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